THE CONVERSATION

Academic rigour, journalistic flair



Philip Thurston/Getty

Australia must double down to meet our climate targets. Our new study shows how

Published: September 14, 2026 6.11am AEST

https://theconversation.com/australia-must-double-down-to-meet-our-climate-targets-our-new-study-shows-how-291668

Within the next five years, global warming is set to breach the 1.5°C threshold at the heart of the 2015 Paris Agreement. This will have dire consequences, from rapidly melting ice sheets to ghostly white coral reefs.

To date, the world has failed to tackle climate change head-on. We must now focus on limiting the magnitude and duration of "overshoot" – how much global temperatures exceed the 1.5°C limit. Every fraction of a degree our planet warms, and every year spent there, will inflict irreversible ecological damage.

A new report, developed by my research team at Climate Analytics and the Potsdam Institute for Climate Impact Research, asks a simple question: what can Australia do to limit this overshoot?

Read more: ***A record-shattering El Niño is almost here. Here's what that means for the world's weather***

A worrying future

Our latest research suggests the world can still limit peak warming to about 1.7°C, reach net zero greenhouse gas emissions in the 2060s, and bring warming back below 1.5°C by 2100. But only if governments take urgent and ambitious action.

Fossil fuels – including oil, coal and gas – account for more than 75% of global greenhouse gas emissions. To curb climate change, we must rapidly transition away from their use. Australia committed to this goal at COP28 in 2023, at COP30 by signing the Belém Declaration and at the Santa Marta conference in April 2026.

In Australia, we've had too many reminders of what's at stake. The horrific Black Summer fires of 2019–20. Repeated record floods, including across Queensland and northern New South Wales following the 2017 Cyclone Debbie. Not to mention the mass bleaching of the Great Barrier, Ningaloo and Scott reefs.

Now, a potentially record-shattering El Niño is forming. Research suggests it could temporarily push global warming towards 1.7°C or even 1.8°C in 2027, likely bringing extreme heat and drought conditions to Australia.

Our new report shows Australia can realistically meet our climate targets. But that demands we urgently adopt an ambitious emissions pathway, built on the Paris Agreement.

Australian firefighters are increasingly having to prepare for severe bushfires. Aston Brown/Getty

The path forward

A Paris-aligned emissions pathway would require Australia to transform its energy system.

Fossil fuels

Fossil fuels supply about 90% of our country's raw energy and make up a large share of our greenhouse gas emissions. Australia is also one of the world's biggest fossil fuel exporters, producing roughly 4.5% of global carbon dioxide emissions.

To achieve our suggested pathway, Australia's net emissions would need to fall 55% below 2005 levels – the year on which Australia's national targets are based – by 2030 and reach net zero by 2050.

But under current policies, Australia will barely meet its 2030 target of a 43% reduction. Worryingly, this means we will likely arrive at 2050 with 245 million tonnes of residual emissions still to be offset – more than half of today's net emissions. Australia would also need its coal and liquefied natural gas production to drop by about 80% and 25%, respectively, by 2035. Coal exports would have to fall by 95% before 2040.

Fossil fuel demand reduces almost to zero by 2050 under the 'Australian Fossil Fuel Phaseout' (FFPO) scenario

Comparing total primary energy demand from fossil fuels in petajoules (PJ) for the FPPO scenario and Current Policy Projections (CPP) scenario. Dotted lines indicate the specific fossil fuel demand under FPPO.

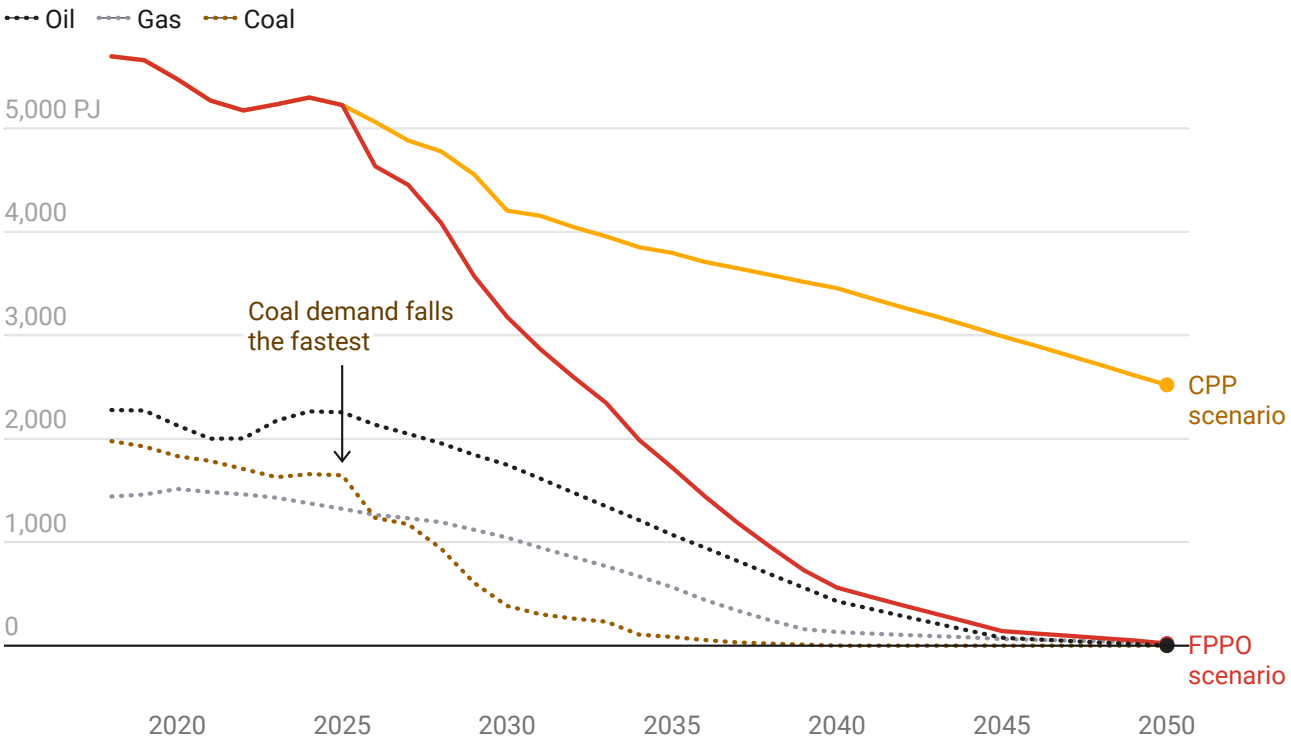


Chart: The Conversation • Source: Climate Analytics: A fossil-free Australia • Get the data • Embed • Download image • Created with Datawrapper

Renewables

Decarbonisation – switching from fossil fuels to renewable energy sources such as wind and solar – is critical to staying below the 1.5°C limit.

To achieve this, electricity must make up 50% of Australia's national energy demand by 2035, up from 25% in 2025. However, current policies would only deliver 33% by 2035, so the 35% by 2035 electrification goal Australia and Turkey are putting forward at this year's COP31 summit is simply business as usual.

Sheep graze around solar panels at the Dubbo Solar Hub, west of Sydney. Gregory Plesse/Getty

Policy must keep up

Achieving these reductions would require major policy changes.

For starters, the current non-binding target of 82% renewables by 2030 – which Australia is likely to miss according to the Australian Energy Market Operator – must be strengthened and made binding. This would make it possible to speed up the renewables rollout, potentially reaching 96% renewables by 2035.

The Safeguard Mechanism – a federal government policy that sets legally binding pollution limits on industrial facilities – must be properly enforced. In 2025, industrial emissions appeared to fall by just 2.3% – less than half the 4.9% annual decline needed. What's more, data from the 196 facilities covered by the Safeguard Mechanism in both 2023–24 and 2024–25 shows their combined emissions dropped by just 0.4%. That's because the mechanism effectively offers companies unlimited offsets that decrease their net emissions.

The moment to fix this is the 2026–27 Safeguard Mechanism review, which is currently receiving stakeholder feedback.

In the oil-dominated transport sector, there are calls for the government to end new petrol and diesel car sales by 2035. However, it has no current plan or policies to electrify heavy vehicle freight. Emissions from mining – one of the few industries where fossil fuel use is still rising – could be pulled down to zero by 2045 if we replace diesel with electricity in trucks, equipment and processing. Scrapping the diesel rebate and other fossil fuel subsidies would also help.

Read more:** ***If all 2030 climate targets are met, the planet will heat by 2.7°C this century. That's not OK

Walking the talk

An Australian fossil fuel phase-out would be a signal of genuine leadership, particularly among our Pacific neighbours already grappling with rapidly rising seas.

Yet, the Albanese government has approved 37 new, expanded or extended fossil fuel projects since coming to power. It has 42 more in the pipeline for approval.

This goes directly against the International Court of Justice which, in its 2025 advisory opinion on climate change, found licensing new fossil fuel projects can constitute an "internationally wrongful act".

Australia is set to lead talks at COP31 in November and has vowed to continue the push for a fossil fuel transition. However, we cannot demand that others respect climate targets we ourselves are failing to meet.

We already have a pathway to keep global warming below 1.5°C. What's missing is the political will to take it.

The Pacific nation of Tuvalu is vulnerable to sea level rise. Mario Tama/Getty

Bill Hare
Adjunct Professor of Energy, Murdoch University

DOI

https://doi.org/10.64628/AA.64nkvahff